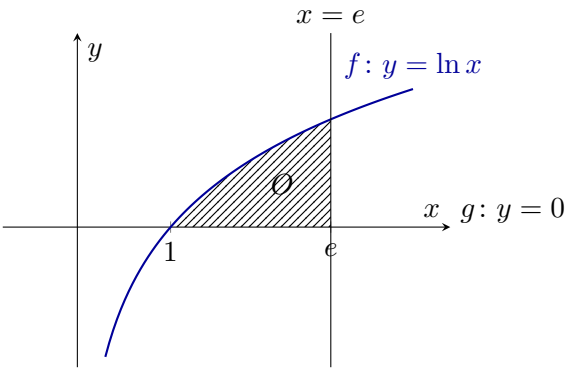
Vypočti:

$$\iint_O y\,\mathrm{d}x\,\mathrm{d}y, \qquad \text{kde } O \text{ je ohraničená křivkami: } \quad y=\ln x, \quad y=0, \quad x=e$$

1. obrázek



Průsečík $f=g$:

$$\begin{aligned} \ln_e x &= 0 \\ e^0 &= x \\ x &= 1 \end{aligned}$$

2. $\boxed{1 \le x \le e}$ zde jsou meze pouze čísla, proto TO MUSÍM dát vně!

$0 \le y \le \ln x$

$$\int_1^e \left(\int_0^{\ln x} y\,\mathrm{d}y \right) \mathrm{d}x = \int_1^e \left[\frac{y^2}{2} \right]_0^{\ln x} \mathrm{d}x = \int_1^e \left(\frac{\ln^2 x}{2} - 0 \right) \mathrm{d}x \overset{*}{=} \left[\frac{1}{2} x \ln^2 x - x \ln x + x \right]_1^e =$$

$$= \frac{1}{2} e \underbrace{(\ln e)^2}_{1} - e \underbrace{\ln e}_{1} + e - \left(\frac{1}{2} \cdot 1 \underbrace{(\ln 1)^2}_{0} - 1 \cdot \underbrace{\ln 1}_{0} + 1 \right) = \frac{1}{2} e - \cancel{e} + \cancel{e} - 1 = \boxed{\frac{e}{2} - 1}$$

∗ Pomocný výpočet:

$$\int \frac{(\ln x)^2}{2}\,\mathrm{d}x = \frac{1}{2} \int (\ln x)^2 \cdot 1\,\mathrm{d}x = \left| \begin{array}{lll} u = (\ln x)^2 & \to & u' = 2 \ln x \cdot \frac{1}{x} \\ v' = 1 & \to & v = x \end{array} \right| =$$

$$= \frac{1}{2} \left(\ln^2 x \cdot x - \int 2 \ln x \frac{1}{\cancel{x}} \cdot \cancel{x}\,\mathrm{d}x \right) = \left| \begin{array}{lll} u = \ln x & \to & u' = \frac{1}{x} \\ v' = 1 & \to & v = x \end{array} \right| =$$

$$= \frac{1}{2} \left(x \ln^2 x - 2 \cdot \left(\ln x \cdot x - \int \frac{1}{\cancel{x}} \cdot \cancel{x}\,\mathrm{d}x \right) \right) = \frac{1}{2} x \ln^2 x - x \ln x + x$$